



SPRING CLOUD CHEAT SHEET

Core setup, initialization, operations & integration quick reference

PART 1: SETUP & CORE

01. SETUP & BOM

Installation

```
<dep>spring-cloud-dependencies BOM</dep>
spring-cloud-starter-config
spring-cloud-starter-netflix-eureka-client
spring-cloud-starter-gateway
```

Match Spring Cloud release train to Spring Boot version

04. SERVICE DISCOVERY (EUREKA)

Workflow

Server: @EnableEurekaServer

```
eureka.instance.hostname=eureka-server
```

Client: @EnableEurekaClient (or auto if dep present)

```
eureka.client.serviceUrl.defaultZone=http://...
spring.application.name=my-service (= service...
```

Clients fetch registry every 30s by default

02. CONFIG SERVER

Architecture

```
@EnableConfigServer @SpringBootApplication
spring.cloud.config.server.git.uri=https://...
spring.cloud.config.server.git.search-paths=c...
GET /{application}/{profile} returns merged c...
```

Config stored in git repo as {app}-{profile}.yml

05. API GATEWAY

CRUD API

```
spring.cloud.gateway.routes:
- id: user-service
uri: lb://USER-SERVICE
predicates:
- Path=/api/users/**
filters:
- StripPrefix=1
```

lb:// = client-side load balancing

03. CONFIG CLIENT

YAML / Config

```
spring.config.import=optional:configserver:
spring.cloud.config.uri=http://config-server:...
spring.cloud.config.fail-fast=true
@RefreshScope on beans to reload on /actuator...
spring.cloud.bus (Kafka/RabbitMQ) for broadca...
```

06. CIRCUIT BREAKER (RESILIENCE4J)

Integration

```
@CircuitBreaker(name="userSvc", fallbackMetho...
@Retry(name="userSvc")
@TimeLimiter(name="userSvc")
resilience4j.circuitbreaker.instances.userSvc
.slidingWindowSize=10
.failureRateThreshold=50
```



SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

07. FEIGN CLIENT

Hardening

```
@EnableFeignClients
@FeignClient(name="user-service")
interface UserClient {
    @GetMapping("/users/{id}")
    User getUser(@PathVariable Long id);
}
```

Integrates with Eureka and Resilience4j

10. SPRING CLOUD BUS

Unit Tests

```
<dep>spring-cloud-starter-bus-kafka</dep>
POST /actuator/busrefresh triggers all clien...
```

Propagates config refresh via message broker

```
spring.cloud.bus.enabled=true
```

Requires Kafka or RabbitMQ broker

08. LOAD BALANCING

Observability

```
<dep>spring-cloud-starter-loadbalancer</dep>
@LoadBalanced RestTemplate template
template.getForObject("http://USER-SERVICE/", ...
WebClient.Builder with @LoadBalanced
```

Round-robin by default; pluggable strategy

11. RETRY & RATE LIMITING

Commands

```
@Retry(name="user", fallbackMethod="fb")
resilience4j.retry.instances.user
.maxAttempts=3
.waitDuration=1s
```

Rate limiter: RequestRateLimiter filter in Gateway

```
redis-rate-limiter.replenishRate=10
```

09. DISTRIBUTED TRACING

Optimization

```
<dep>spring-cloud-starter-sleuth</dep>
<dep>zipkin-brave-spring-cloud-sleuth</dep>
```

MDC populated with traceId, spanId

```
spring.zipkin.base-url=http://zipkin:9411
spring.sleuth.sampler.probability=1.0
```

12. COMMON GOTCHAS

Warning

Release train version mismatch causes NoSuchMethod errors
Feign client needs @EnableFeignClients on main class
Config Server needs to start before clients (health check)
Ribbon deprecated use spring-cloud-starter-loadbalancer
Sleuth auto-adds trace headers; verify sampling rate in prod